A Breif Epilogy in Medical Sciences - Extended First Edition

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Contents

Preface

We have chosen the name to be Medical Sciences , as this the name for the studies at Intermediate level.

The Human Body

Part I

The Brain

At the back side of the brain is the grey brain. It is extremely small in size compared to the rest of the brain. It helps in memorization prosess. Apart from this the brain has many short functional parts which are said to perform various functions according to modern studies. In old times in Indian texts there is a mention of talu on the skull top which was called to be one of the important functionaries in indi treatement. Then there are the pudpudis on the sided of the brain, extremely soft.

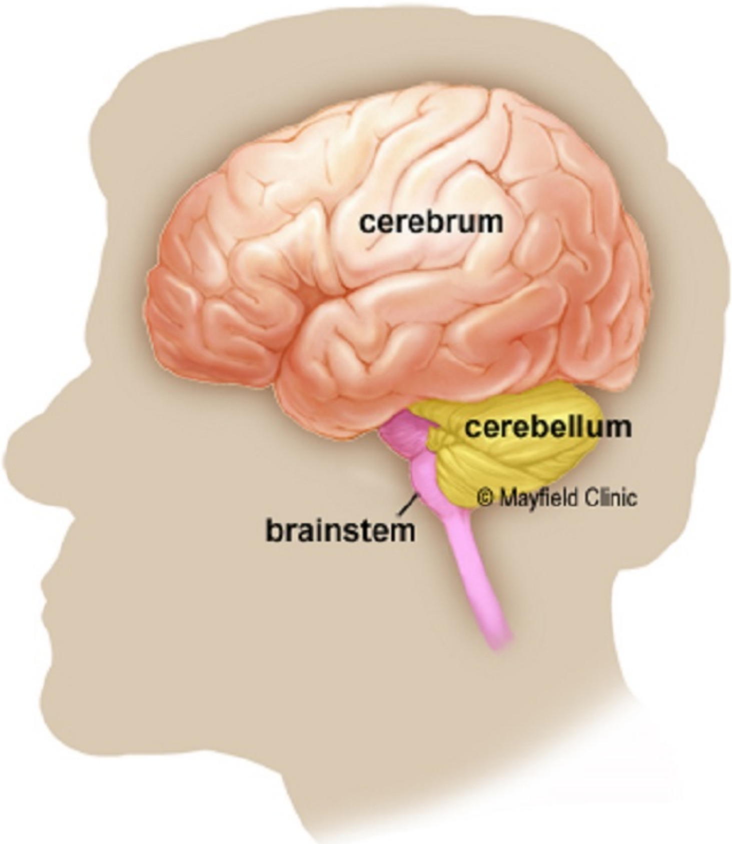
Chapter 1

Cerebrum, Cerebellum, and Brainstem

Cerebrum: is the largest part of the brain and is composed of right and left hemispheres. It performs higher functions like interpreting touch, vision and hearing, as well as speech, reasoning, emotions, learning, and fine control of movement.

Cerebellum: is located under the cerebrum. Its function is to coordinate muscle movements, maintain posture, and balance.

Brainstem: acts as a relay center connecting the cerebrum and cerebellum to the spinal cord. It performs many automatic functions such as breathing, heart rate, body temperature, wake and sleep cycles, digestion, sneezing, coughing, vomiting, and swallowing.



Part II

The Heart

Usually you get to hear about operations when the valve of a person's heart is replaced. Then modern medical studies say that there are four chambers in heart. Then there is the Aorta , which passes through the heart.

Part III

The Respiratory Organs

We breathe through our mouth, then there is a stoppage in the neck which blocks any eatables to enter the respiratory pipe when one is eating. Then there are the lungs which filter the air we have breathed and supply clean oxygen in the arteries.

Part IV

The Kidneys

They filter the blood and produce all the liquid excretory wastes passed out through the urinal passing through the bladder. Any of the Kidney has two chambers.

Part V

The Bones

Chapter 2

Dental Bones (as a Special Case)

A normal person has 32 teeth comprising of frontal teeth , molars , canines etc.

Part VI

Ear, Nose and Throat (ENT)

We have two ears (which have eardrums), a nose with two hoses and hair and a throat with tonsils.

Part VII

The Skin

It comprisis of two layers of epidermis and on the outside covered with hair at most parts in men.

Part VIII

Sexual Organs

Men have a penis, women have a vagina as against this. Then men have a pair of testicles. On the other side women have egg producing sachets and a womb too.

Part IX

Maternity (Amongst Women)

Women can get pregnant. It is a highly fragile state in a women's life while giving birth and some women with delivery experience claim that they have got second life.

A Plant or a Tree

Part X

Plant Anatomy

Chapter 3

The Stem

It draws water from the roots along with other nutrients and provides to the rest of the upper plant parts. It has a thick outer covering which has celluloid in the interior to provide a firm state to it.

Chapter 4

The Root

The roots draw water from soil through sub roots or tentacles. Apart from it they draw nutrients too. Some of the roots like potato have nitrogen fixing bacteria too.

Chapter 5

The Branches

Later.

Chapter 6

The Leaves

They do photosynthesis i.e. food production for the plant and oxygen too.

Chapter 7

The Flower

It's the reproductive organ of the plant . So, if a plant has many flowers , we can say that it has opened multiple sites for reproduction , a phenominon completely unseen in humans.

Chapter 8

The Fruit

Fruits can be poisionous or eatable. They contain seeds too to create the next generation of the same plant . (unless the plant is modified by branch “pyonde”)

Part XI

Plants which can move

Chapter 9

The Chui-Mui

It can shy away if somebody touches it.

Chapter 10

Plants which can consume an animal(say a small insect) as food

Insectivourous plants have a flower with a cup shape and a lid like opening . So an unwary insect which comes to take pollen gets struck and killed in return instead of taking pollen. It's possible that these plants can be bisexual or finding other ways of pollination as insects could not take pollen .

Part XII

Plants used as food by humans or animals

It comprises of all major plants like spinach.

Chapter 11

Wheat

The seeds are consumed by humans while the bark by animals as hay.

Chapter 12

Sugarcane

The stem can be used for direct consumption or it's juice can produce sugar , gur , jaggery etc.

Part XIII

Trees bearing Fruits and Plants bearing vegetables

Chapter 13

Trees bearing Fruits

13.1 Apple

13.2 Orange

Chapter 14

Plants bearing vegetables

Many plants bear vegetables like Cucumber, tomato (disputed between a vegetable and a fruit) , potato etc.

Part XIV

Allergic Plants and Trees

Chapter 15

The Congress Grass

It can cause minor allergy. Although it has nothing to do with congress party it seems. The name probably has been kept in vein. It's found in abundance in Chandigarh.

Part XV

Plants having products of medicinal value

Chapter 16

Plant sources for special Organic Chemicals

Chapter 17

Under Ayurvedic System

Various plants were claimed to be of great medicinal value in the olden times when modern equipments and chemical methods were not present. Baba ramdev and his team are struggling hard to get ayurvedic a place in modern world. Russels Peters have made mockery of this system by calling ikjot's possible sister , a herbaljot.

Ayurvedic medicine is made of those plants which can't be used directly as food and have some fast action in body to maybe counter a disease they say.

Part XVI

Plants having products worthy in market

Part XVII

Plants which can be grafted to the Bonzai

Part XVIII

Plants or Trees having products valued in Narcotics

Bhang (Ganja), Hasheesh etc have been known to be plant oriented narcotics.

Other Animals

Part XIX

Mammals

Part XX

Insects

Part XXI

Reptiles

Part XXII

Aquatic Animals

Part XXIII

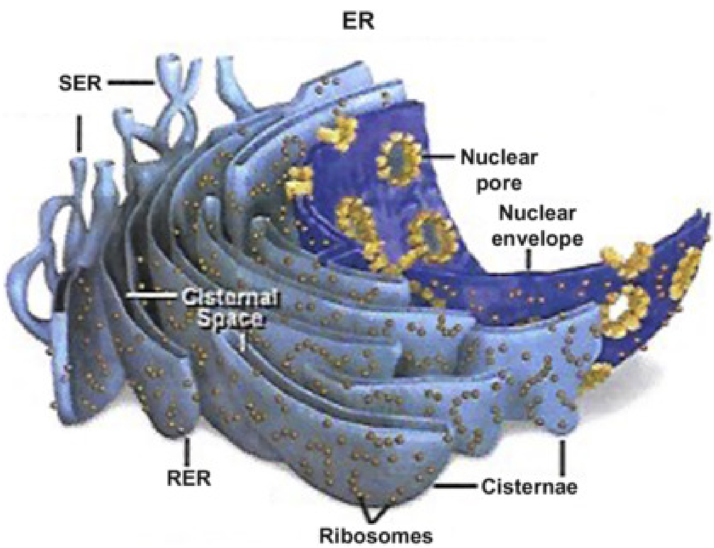
Animals Intermediate between these Classifications

Studying Micro-organisms

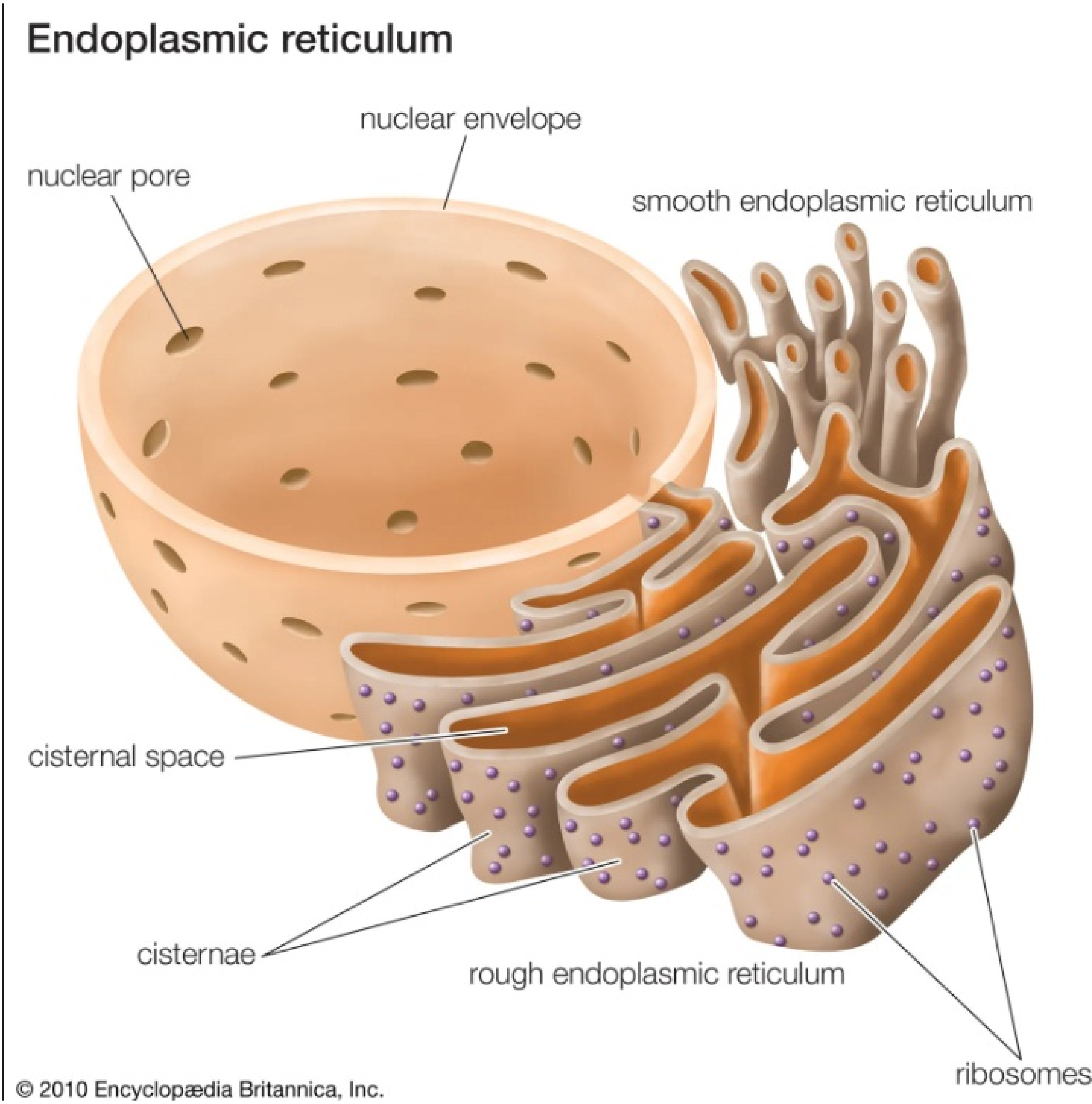
Chapter 18

Cell Organalles

18.1 ER {Endoplasmic Reticulum}



The endoplasmic reticulum (ER) is a network of interconnected membranes within eukaryotic cells, playing a crucial role in protein and lipid synthesis, modification, and transport. It exists in two forms: rough endoplasmic reticulum (RER), which is studded with ribosomes and involved in protein synthesis, and smooth endoplasmic reticulum (SER), which lacks ribosomes and is involved in lipid synthesis and other functions.



• Structure:

The ER is a continuous network of membrane-bound tubules and flattened sacs (cisternae) that extends from the nuclear membrane throughout the cytoplasm.

• Rough Endoplasmic Reticulum (RER):

The RER is characterized by the presence of ribosomes on its outer surface, which are responsible for protein synthesis. Proteins synthesized on the RER are often destined for other organelles, the cell membrane, or secretion outside the cell.

• Smooth Endoplasmic Reticulum (SER):

The SER lacks ribosomes and is involved in a variety of functions, including lipid synthesis (steroid hormones, phospholipids), detoxification of drugs and toxins, and the release of calcium ions for muscle contraction and nerve function.

• Functions:

The ER performs various functions, including:

• Protein Synthesis: The RER is the primary site of protein synthesis, especially for proteins destined for secretion or insertion into the cell membrane.

• Lipid Synthesis: The SER is involved in the synthesis of various lipids, including steroids, phospholipids, and cholesterol.

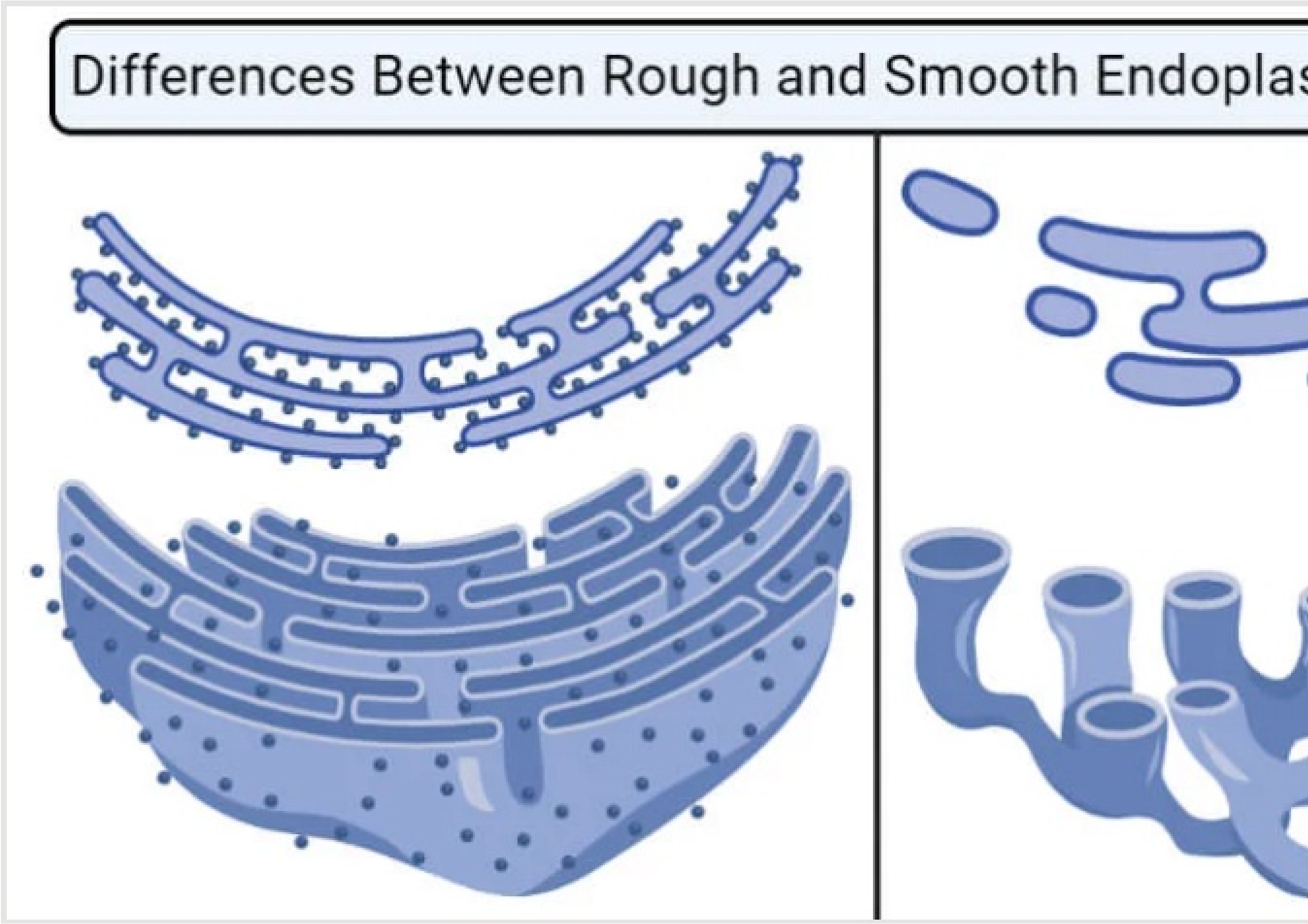
• Detoxification: The SER can help detoxify harmful substances in the body.

• Calcium Storage and Release: The ER can store and release calcium ions, which are crucial for muscle contraction and nerve function.

• Protein Folding and Modification: The ER plays a role in protein folding and modification, ensuring proteins are properly shaped and functional.

Relationship to Other Organelles:

The ER is connected to the nuclear envelope and other organelles, including the Golgi apparatus, allowing for the efficient movement and modification of proteins and lipids.



18.2 RER {Rough Endoplasmic Reticulum}

The rough endoplasmic reticulum (RER) is a network of membrane-bound sacs and tubules found in the cytoplasm of eukaryotic cells. It's characterized by ribosomes attached to its surface, giving it a "rough" appearance. The RER plays a crucial role in protein synthesis, folding, and modification, especially for proteins destined for secretion or incorporation into other cell membranes.

Here's a more detailed explanation:

Structure:

• Membrane Sheets:

The RER consists of flattened, interconnected sacs called cisternae, which extend throughout the cytoplasm, often near the nucleus.

• Ribosomes:

Ribosomes, the protein synthesis machinery, are attached to the outer surface of the RER membrane.

• Connection to the Nuclear Envelope:

The RER is physically connected to the nuclear envelope, which surrounds the nucleus, allowing for efficient protein transport.

Function:

• Protein Synthesis:

Ribosomes on the RER translate mRNA into proteins, with proteins destined for secretion or incorporation into the cell membrane being synthesized here.

- Protein Folding and Modification:

The RER provides a space for newly synthesized proteins to fold into their correct three-dimensional shape and undergo modifications, such as glycosylation (addition of sugar molecules).

- Protein Transport:

The RER is involved in transporting synthesized proteins to other cellular compartments, such as the Golgi apparatus, which further processes and packages proteins.

- Quality Control:

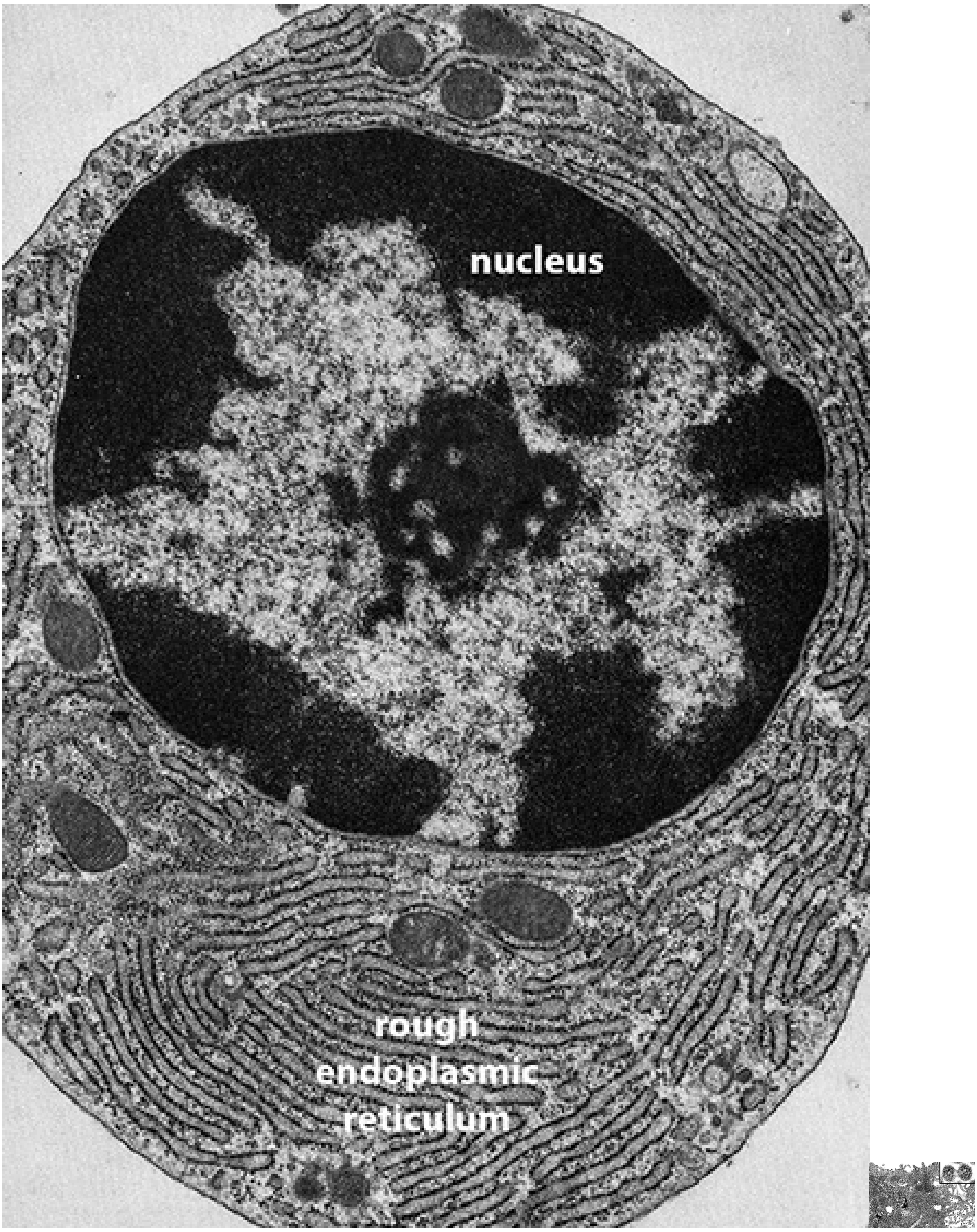
The RER also plays a role in protein quality control, ensuring that only properly folded proteins are sent on to other destinations.

Key Differences from Smooth Endoplasmic Reticulum (SER):

- Ribosomes: The RER has ribosomes attached to its surface, while the SER does not.

- Function: RER primarily focuses on protein synthesis, while SER is involved in lipid synthesis, detoxification, and calcium storage.

In essence, the RER is a crucial cellular organelle responsible for protein synthesis, modification, and transport, particularly for proteins destined for export from the cell or incorporation into the cell's membranes.


nucleus
rough
endoplasmic
reticulum

Part XXIV

Their Importance in Medicine

Chapter 19

Microbial Wastes

Part XXV

Bacteria

Part XXVI

Viruses

Part XXVII

Fungus

Bio-studies

Part XXVIII

Study of Excreta

Chapter 20

Water Purification Processes

Part XXIX

Modern new Dimension

Chapter 21

Plant residue